

[Continuation and conclusion of Memoir No. 146 (“A Memoir on Curves of the Third Order”), Articles Nos. 35–41. The opening of the memoir, together with Articles Nos. 1–34, occupies PDF pp. 401–425 of the present volume.]

(From p. 409 of the printed memoir, end of Article No. 35.) Adding these values and completing the reduction, we find

$$(\xi^3 + \eta^3 + \zeta^3) - (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) = -\xi^6 - \eta^6 - \zeta^6 + 3\xi^2\eta^2\zeta^2 + 2\xi^2\eta^4 + 2\xi^4\eta^2 + \dots \\ + 18l\xi^2\eta^2\zeta^2 + 12l^2(\xi^4 + \eta^4 + \zeta^4)\xi\eta\zeta + 8l^3(\xi^2\eta^2 + \eta^2\zeta^2 + \zeta^2\xi^2),$$

and we have also

$$\xi\eta\zeta [A(x^3 + y^3 + z^3) + Bxyz] = (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) \\ - A(\xi^6 + \eta^6 + \zeta^6) \\ + (12l^2A + lB)(\xi^4 + \eta^4 + \zeta^4)\xi\eta\zeta \\ + (18lA + (1 + 8l^3)B)\xi^2\eta^2\zeta^2 \\ + (\text{further symmetric terms in the } \xi, \eta, \zeta).$$

The coefficient of $\xi^4\eta^2 + \eta^4\zeta^2 + \zeta^4\xi^2$ on the right-hand side will vanish if

$$A + l(1 + 2l^3) + l^2B = 0,$$

that is, if $A = -l$, $B = \frac{1}{l}(1 + 2l^3)$ (or, what is the same thing, $l^2A = -l^3$); and substituting these values, we obtain

$$l\{l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz\} - (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) \\ = -l(\xi^6 + \eta^6 + \zeta^6) \\ + (-l + 4l^4)(\xi^4\eta^2 + \eta^4\zeta^2 + \zeta^4\xi^2 + \xi^2\eta^4 + \eta^2\zeta^4 + \zeta^2\xi^4) \\ + (-l + 8l^4 - 16l^7)\xi^2\eta^2\zeta^2,$$

or, what is the same thing,

$$l\{l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz\} = -(\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) + \{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\}^2.$$

Hence the left-hand side vanishes in virtue of the relation between ξ, η, ζ , or we have

$$l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0,$$

which proves the theorem.

36. Suppose that (X, Y, Z) are the coordinates of a point of the Hessian, and let (P, Q, R) be the coordinates of the point in which the tangent to the Hessian at the point (X, Y, Z) again meets the Hessian, or, what is the same thing, the satellite point in regard to the Hessian of the tangent at (X, Y, Z) . And consider the conic

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy),$$

which is the first or conic polar of the point (X, Y, Z) in respect of the cubic. The polar (in respect to this conic) of the point (P, Q, R) will be

$$\xi x + \eta y + \zeta z = 0,$$

where

$$\xi = PX + l(RY + QZ), \\ \eta = QY + l(PZ + RX), \\ \zeta = RZ + l(QX + PY);$$

or putting for (P, Q, R) their values,

$$\begin{aligned}\xi &= (Y^3 - Z^3)(X^2 - lYZ), \\ \eta &= (Z^3 - X^3)(Y^2 - lZX), \\ \zeta &= (X^3 - Y^3)(Z^2 - lXY);\end{aligned}$$

and if from these equations and the equation of the Hessian we eliminate (X, Y, Z) , we shall obtain the equation in line coordinates of the curve which is the envelope of the line $\xi x + \eta y + \zeta z = 0$. We find, in fact,

$$\begin{aligned}\xi^3 + \eta^3 + \zeta^3 &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{-l^2(X^3 + Y^3 + Z^3) - 3l(X^3 + Y^3 + Z^3)XYZ + 9l^2X^2Y^2Z^2 + (1 - 4l^3)(Y^2Z^2 + Z^2X^2 + X^2Y^2)\} \\ \xi\eta\zeta &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{-l(X^3 + Y^3 + Z^3)XYZ + (1 - l^3)X^2Y^2Z^2 - l(Y^2Z^2 + Z^2X^2 + X^2Y^2)\}\end{aligned}$$

and thence recollecting that

$$HU = l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ,$$

we find

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = -(Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3)(HU)^2,$$

and the equation of the envelope is

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

which is therefore the Pippian. We have thus the theorem:

The envelope of the polar of the satellite point in respect to the Hessian of the tangent at any point of the Hessian, such polar being in respect of the conic which is the first or conic polar of the point of the Hessian in respect of the cubic, is the Pippian.

Article Nos. 37 to 40. Investigations and theorems relating to the first or conic polar of a point of the cubic.

37. The investigations next following depend on the identical equations

$$\begin{aligned}&\{\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)\} \\ &\quad \times [-XYZ(x^3 + y^3 + z^3) + (X^3 + Y^3 + Z^3)xyz] \\ &= \{X(x^2 + 2lyz) + Y(y^2 + 2lxz) + Z(z^2 + 2lxy)\} \\ &\quad \times [X(Y^2 - Z^2)(\gamma y - \beta z) + Y(Z^2 - X^2)(\alpha z - \gamma x) + Z(X^2 - Y^2)(\beta x - \alpha y)] \\ &\quad + \{x(X^2 + 2lYZ) + y(Y^2 + 2lZX) + z(Z^2 + 2lXY)\} \\ &\quad \times [-(\alpha YZ + \beta ZX + \gamma XY)(Xx^2 + Yy^2 + Zz^2) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)(Xyz + Yzx + Zxy)],\end{aligned}$$

which is easily verified.

I represent the equation in question by

$$KT = WL + P\Theta;$$

then considering (x, y, z) as current coordinates, and (X, Y, Z) and (α, β, γ) as the coordinates of two given points Σ and Ω , we shall have $U = 0$ the equation of the cubic, $W = 0$ the equation of the first or conic polar of Σ with respect to the cubic, $P = 0$ the equation of the second or line polar of Σ with respect to the cubic. The equation $T = 0$ is that of a syzygetic cubic passing through the point Σ : the coordinates of the satellite point in respect to this syzygetic cubic of its tangent at Σ are

$$X(Y^3 - Z^3) : Y(Z^3 - X^3) : Z(X^3 - Y^3);$$

and calling the point in question Σ' , then $L = 0$ is the equation of a line through the points Σ', Ω . The equation $\Theta = 0$ is that of a conic, viz. the first or conic polar of Σ with respect to a certain syzygetic cubic

$$-2(\alpha YZ + \beta ZX + \gamma XY)(x^3 + y^3 + z^3) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)xyz = 0,$$

depending on the points Σ , Ω , or, what is the same thing, the conic $\Theta = 0$ is a properly selected conic passing through the points of intersection of the first or conic polars of Σ with respect to any two syzygetic cubics; and lastly, K is a constant coefficient. The equation expresses that the points of intersection of

$$(W = 0, \Theta = 0), \quad (W = 0, P = 0), \quad (L = 0, \Theta = 0), \quad (L = 0, P = 0),$$

lie in the syzygetic cubic $T = 0$.

The left-hand side of the equation may be written

$$\begin{aligned} & -XYZ \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \} (x^3 + y^3 + z^3 + 6lxyz) \\ & + xyz \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \} (X^3 + Y^3 + Z^3 + 6lXYZ); \end{aligned}$$

and it may be remarked also that we have

$$-3XYZ \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \}$$

equal identically to

$$\begin{aligned} & [X(Y^2 - Z^2)(\gamma Y - \beta Z) + Y(Z^2 - X^2)(\alpha Z - \gamma X) + Z(X^2 - Y^2)(\beta X - \alpha Y)] \\ & - (\alpha YZ + \beta ZX + \gamma XY)(X^3 + Y^3 + Z^3 + 6lXYZ). \end{aligned}$$

Hence if we assume

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0,$$

the equation will take the form

$$KU = WL + P\Theta,$$

where the constant coefficient K may be expressed under the two different forms

$$\begin{aligned} K &= -XYZ [\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)] \\ &= \frac{1}{3} [X(Y^2 - Z^2)(\gamma Y - \beta Z) + Y(Z^2 - X^2)(\alpha Z - \gamma X) + Z(X^2 - Y^2)(\beta X - \alpha Y)], \end{aligned}$$

and W , L , P , Θ have the same values as before. In the present case the point Σ is a point of the cubic: the equation $W = 0$ represents the first or conic polar of the point in question, and the equation $P = 0$ its second or line polar, which is also the tangent of the cubic. The line $L = 0$ is a line joining the point Ω with the satellite point of the tangent at Σ , or dropping altogether the consideration of the point Ω , is an arbitrary line through the satellite point: the first or conic polar of Σ meets the cubic twice in the point Σ , and therefore also meets it in four other points; the conic $\Theta = 0$ is a conic passing through these four points, and completely determined when the particular position of the line through the satellite point is given. And, as before remarked, $\Theta = 0$ is a conic passing through the points of intersection of the first or conic polars of Σ with respect to any two syzygetic cubics. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at this point, and besides meets it in four other points; the four points in question are the points in which the first or conic polar of the given point in respect of the cubic is intersected by the first or conic polar of the same point in respect to any syzygetic cubic whatever.

38. The analytical result may be thus stated: putting

$$\kappa = \alpha YZ + \beta ZX + \gamma XY, \quad \lambda = \alpha X^2 + \beta Y^2 + \gamma Z^2,$$

or, if we please, considering κ, λ as arbitrary parameters, then the four points lie in the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0,$$

or, what is the same thing, they are the points of intersection of the two conics

$$Xx^2 + Yy^2 + Zz^2 = 0,$$

$$Xyz + Yzx + Zxy = 0.$$

39. Considering the four points as the angles of a quadrangle, it may be shown that the three centres of the quadrangle lie on the cubic. To effect this, assume that the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0$$

represents a pair of lines; these lines will intersect in a point, which is one of the three centres in question. And taking x, y, z as the coordinates of this point, we have

$$\begin{aligned} x^2 : y^2 : z^2 : yz : zx : xy \\ = 4\kappa^2 YZ - \lambda^2 X^2 : 4\kappa^2 ZX - \lambda^2 Y^2 : 4\kappa^2 XY - \lambda^2 Z^2 \\ : \lambda^2 YZ + 2\kappa\lambda X^2 : \lambda^2 ZX + 2\kappa\lambda Y^2 : \lambda^2 XY + 2\kappa\lambda Z^2; \end{aligned}$$

and we may, if we please, use these equations to find the relation between κ, λ . Thus in the identical equation $x^2 \cdot y^2 - (xy)^2 = 0$, substituting for x^2, xy, y^2 their values, and throwing out the factor Z , we find $(4\kappa^2 - \lambda^2)XYZ - \kappa\lambda^2(X^3 + Y^3 + Z^3) = 0$, and thence, in virtue of the equation $X^3 + Y^3 + Z^3 + 6lXYZ = 0$, we obtain

$$4\kappa^2 + 6l\kappa\lambda^2 - \lambda^2 = 0.$$

But the preceding system gives conversely,

$$\begin{aligned} X^2 : Y^2 : Z^2 : YZ : ZX : XY \\ = 4\kappa^2 yz - \lambda^2 x^2 : 4\kappa^2 zx - \lambda^2 y^2 : 4\kappa^2 xy - \lambda^2 z^2 \\ : \lambda^2 yz + 2\kappa\lambda x^2 : \lambda^2 zx + 2\kappa\lambda y^2 : \lambda^2 xy + 2\kappa\lambda z^2. \end{aligned}$$

Hence from the identical relation $X^2 \cdot Y^2 - (XY)^2 = 0$, substituting for X^2, XY, Y^2 their values, and throwing out the factor z , we find $(4\kappa^2 - \lambda^2)xyz - \kappa\lambda^2(x^3 + y^3 + z^3) = 0$, and thence, in virtue of the equation $4\kappa^2 - \lambda^2 = -6l\kappa\lambda^2$, we obtain

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

which shows that the point in question lies on the cubic. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at the point, and meets it besides in four points, which are the angles of a quadrangle the centres of which lie on the cubic. In other words, the quadrangle is an inscribed quadrangle.

40. To find the equations of the three axes of the quadrangle, that is of the lines through two centres.

We have

$$\begin{aligned} (4\kappa^2 YZ - \lambda^2 X^2)x + (\lambda^2 XY + 2\kappa\lambda Z^2)y + (\lambda^2 ZX + 2\kappa\lambda Y^2)z &= 0, \\ (\lambda^2 XY + 2\kappa\lambda Z^2)x + (4\kappa^2 ZX - \lambda^2 Y^2)y + (\lambda^2 YZ + 2\kappa\lambda X^2)z &= 0, \\ (\lambda^2 ZX + 2\kappa\lambda Y^2)x + (\lambda^2 YZ + 2\kappa\lambda X^2)y + (4\kappa^2 XY - \lambda^2 Z^2)z &= 0; \end{aligned}$$

or arranging these equations in the proper form and eliminating $\kappa^2, \kappa\lambda, \lambda^2$, we find

$$\begin{vmatrix} YZx, Z^2y + Y^2z, X(-Xx + Yy + Zz) \\ ZXy, X^2z + Z^2x, Y(-Xx - Yy + Zz) \\ XYz, Y^2x + X^2y, Z(-Xx + Yy - Zz) \end{vmatrix} = 0;$$

or, multiplying out,

$$\begin{aligned} XYZ \{ (Z^2 - Y^2)x^3 + (X^2 - Z^2)y^3 + (Y^2 - X^2)z^3 \} \\ + x^2yZY^2(-2X^2 + Y^2 + Z^2) + zx^2YZ^2(2X^2 - Y^2 - Z^2) \\ + y^2zXZ^2(-2Y^2 + X^2 + Z^2) + xy^2ZX^2(2Y^2 - Z^2 - X^2) \\ + z^2xYX^2(-2Z^2 + X^2 + Y^2) + yz^2XY^2(2Z^2 - X^2 - Y^2) = 0. \end{aligned}$$

We may simplify this result by means of the equation $X^3 + Y^3 + Z^3 + 6lXYZ = 0$, so as to make the left-hand side divide out by XYZ : we thus obtain

$$\begin{aligned} & (Z^3 - Y^3)x^3 + (X^3 - Z^3)y^3 + (Y^3 - X^3)z^3 \\ & + (-3X^2Y - 6lY^2Z)x^2y + (-3Y^2Z - 6lZ^2X)y^2z + (-3Z^2X - 6lX^2Y)z^2x \\ & + (3XY^2 + 6lX^2Z)xy^2 + (3YZ^2 + 6lY^2X)yz^2 + (3ZX^2 + 6lZ^2Y)zx^2 = 0; \end{aligned}$$

or in a different form,

$$\begin{aligned} & (y^3 - z^3)X^3 + (z^3 - x^3)Y^3 + (x^3 - y^3)Z^3 \\ & + (-3x^2y - 6lz^2x)X^2Y + (-3y^2z - 6lx^2y)Y^2Z + (-3z^2x - 6ly^2z)Z^2X \\ & + (3xy^2 + 6lyz^2)YZ^2 + (3yz^2 + 6lx^2z)ZX^2 + (3zx^2 + 6lxy^2)XY^2 = 0, \end{aligned}$$

as the equation of the three axes of the quadrangle.

Article No. 41. Recapitulation of geometrical definitions of the Pippian.

In conclusion, I will recapitulate the different modes of generation or geometrical definitions of the Pippian, obtained in the course of the present memoir. The curve in question is:

1. The envelope of the line joining a pair of conjugate poles of the cubic (see Nos. 2 and 13).
2. The envelope of each line of the pair forming the first or conic polar with respect to the cubic of a conjugate pole of the cubic (see Nos. 2 and 14).
3. The envelope of a line which is the polar of a conjugate pole of the cubic, with respect to the conic which is the first or conic polar of the other conjugate pole in respect to any syzygetic cubic (see Nos. 2 and 9).
4. The locus of the harmonic with respect to a pair of conjugate poles of the cubic of the third point of intersection with the Hessian of the line joining the two conjugate poles (see Nos. 2 and 17).
5. The envelope of a line such that its lineo-polar envelope with respect to the cubic breaks up into a pair of lines (see No. 24).
6. The envelope of a line which meets three conics, the first or conic polars of any three points in respect to the cubic, in six points in involution (see No. 22).
7. The envelope of the second or line polar with respect to the cubic, of a point the locus of which is a certain curve of the sixth order in quadratic syzygy with the cubic and Hessian, viz. the curve $-S \cdot U^2 + (HU)^2 = 0$ (see No. 27).
8. The envelope of a line having for its satellite point a point of the Hessian (see No. 35).
9. The envelope of the polar of the satellite point with respect to the Hessian of the tangent at a point of the Hessian, with respect to the first or conic polar of the point of the Hessian in respect to the cubic (see No. 36).

147.

A MEMOIR ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF AN EQUATION.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 489–499. Received December 18, 1856,—Read January 8, 1857.]

There are contained in a work, which is not, I think, so generally known as it deserves to be, the “Algebra” of Meyer Hirsch [the work referred to is entitled *Sammlung von Beispielen Formeln und Aufgaben aus der Buchstabenrechnung und Algebra*, 8vo. Berlin, 1804 (8 ed. 1853), English translation by Ross, 8vo. London, 1827] some very useful tables of the symmetric functions up to the tenth degree of the roots of an equation of any order. It seems desirable to join to these a set of tables, giving reciprocally the expressions of the powers and products of the coefficients in terms of the symmetric functions of the roots. The present

memoir contains the two sets of tables, viz. the new tables distinguished by the letter (a), and the tables of Meyer Hirsch distinguished by the letter (b); the memoir contains also some remarks as to the mode of calculation of the new tables, and also as to a peculiar symmetry of the numbers in the tables of each set, a symmetry which, so far as I am aware, has not hitherto been observed, and the existence of which appears to constitute an important theorem in the subject. The theorem in question might, I think, be deduced from a very elegant formula of M. Borchardt (referred to in the sequel), which gives the generating function of any symmetric function of the roots, and contains potentially a method for the calculation of the Tables (b), but which, from the example I have given, would not appear to be a very convenient one for actual calculation.

Suppose in general

$$(1, b, c, \dots | 1, x)^n = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots,$$

so that

$$-b = \Sigma \alpha, \quad +c = \Sigma \alpha \beta, \quad -d = \Sigma \alpha \beta \gamma, \quad \&c.,$$

and if in general

$$(pqr \dots) = \Sigma \alpha^p \beta^q \gamma^r \dots,$$

where as usual the summation extends only to the distinct terms, so that e.g. (p^2) contains only half as many terms as (pq) , and so in all similar cases, then we have

$$-b = (1), \quad +c = (1^2), \quad -d = (1^3), \quad \&c.;$$

and the two problems which arise are, first to express any combination $b^p c^q \dots$ in terms of the symmetric functions $(l^\lambda m^\mu \dots)$, and secondly, or conversely, to express any symmetric function $(l^\lambda m^\mu \dots)$ in terms of the combinations $b^p c^q \dots$.

It will conduce materially to brevity if $1^p 2^q \dots$ be termed the partition belonging to the combination $b^p c^q \dots$; and in like manner if $l^\lambda m^\mu \dots$ be termed the partition belonging to the symmetric function $(l^\lambda m^\mu \dots)$, and if the sum of the component numbers of the partition is termed the weight.

Consider now a line of combinations corresponding to a given weight, e.g. the weight 4, this will be

$$\begin{array}{cccccc} e & bd & c^2 & b^2 c & b^4 & \text{(line)} \\ 4 & 13 & 2^2 & 1^2 2 & 1^4 & \end{array}$$

where I have written under each combination the partition which belongs to it, and in like manner a column of symmetric functions of the same weight, viz.

$$\begin{array}{c} (4) \quad \text{(column)} \\ (31) \\ (2^2) \\ (21^2) \\ (1^4), \end{array}$$

where, as the partitions are obtained by simply omitting the $()$, I have not separately written down the partitions.

It is at once obvious that the different combinations of the line will be made up of numerical multiples of the symmetric functions of the column; and conversely, that the symmetric functions of the column will be made up of numerical multiples of the combinations of the line; but this requires a further examination. There are certain restrictions as to the symmetric functions which enter into the expression of the combination, and conversely, as to the combinations which enter into the expression of the symmetric function. The nature of the first restriction is most clearly seen by the following Table:

No. of Parts.	Greatest Part.	Comb. with Combinations.	Their Partitions.	Contain Multiples of the Symmetric Functions.	Greatest Part does not exceed.	No. of Parts not less than.
1	4	e	4	(1^4)	1	4
2	3	bd	13	$(1^4), (21^2)$	2	3
2	2	c^2	2^2	$(1^4), (21^2), (2^2)$	2	2
3	2	$b^2 c$	$1^2 2$	$(1^4), (21^2), (2^2), (31)$	3	2
4	1	b^4	1^4	$(1^4), (21^2), (2^2), (31), (4)$	4	1

Thus, for instance, the combination bd (the partition whereof is 13) contains multiples of the two symmetric functions (1^4) , (21^2) only. The number of parts in the partition 13 is 2, and the greatest part is 3. And in the partitions (1^4) , (21^2) the greatest part is 2, and the number of parts is not less than 3. The reason is obvious: each term of the developed expression of bd must contain at least as many roots as are contained in each term of d , that is 3 roots, and since the coefficients are linear functions in respect to each root, the combination bd cannot contain a power higher than 2 of any root. The reasoning is immediately applied to any other case, and we obtain

First Restriction.—A combination $b^p c^q \dots$ contains only those symmetric functions $(l^\lambda m^\mu \dots)$, for which the greatest part does not exceed the number of parts in the partition $1^p 2^q \dots$, and the number of parts is not less than the greatest part in the same partition.

Consider a partition such as $1^2 2$, then replacing each number by a line of units thus,

$$\begin{array}{c} 1 \\ 1 \\ 11 \end{array}$$

and summing the columns, we obtain a new partition 31, which may be called the conjugate¹ of $1^2 2$. It is easy to see that the expression for the combination $b^2 c$ (for which the partition is $1^2 2$) contains with the coefficient unity, the symmetric function (31), the partition whereof is the conjugate of $1^2 2$. In fact $b^2 c = (-\Sigma\alpha)^2(\Sigma\alpha\beta)$, which obviously contains the term $1\alpha^3\beta$, and therefore the symmetric function with its coefficient +1 (31); and the reasoning is general, or

Theorem. A combination $b^p c^q \dots$ contains the symmetric function (partition conjugate to $1^p 2^q \dots$) with the coefficient unity, and sign + or – according as the weight is even or odd.

Imagine the partitions arranged as in the preceding column, viz. first the partition into one part, then the partitions into two parts, then the partitions into three parts, and so on; the partitions into the same number of parts being arranged according to the magnitude of the greatest part (the greatest magnitude first), and in case of equality according to the magnitudes of the next greatest part, and so on (for other examples, see the outside column of any one of the Tables). The order being thus completely defined, we may speak of a partition as being prior or posterior to another. We are now able to state a second restriction as follows.

Second Restriction.—The combination $b^p c^q \dots$ contains only those symmetric functions which are of the form (partition not prior to the conjugate of $1^p 2^q \dots$).

The terms excluded by the two restrictions are many of them the same, and it might at first sight appear as if the two restrictions were identical; but this is not so: for instance, for the combination bd^2 , see Table VII (a), the term (41^2) is excluded by the first restriction, but not by the second; and on the other hand, the term $(3^2 1)$, which is not excluded by the first restriction, is excluded by the second restriction, as containing a partition $3^2 1$ prior in order to 32^2 , which is the partition conjugate to 13^2 , the partition of bd^2 . It is easy to see why bd^2 does not contain the symmetric function $(3^2 1)$; in fact, a term of $(3^2 1)$ is $(\alpha^3\beta^3\gamma)$, which is obviously not a term of $bd^2 = (-\Sigma\alpha)(\Sigma\alpha\beta\gamma)^2$; but I have not investigated the general proof.

I proceed to explain the construction of the Tables (a). The outside column contains the symmetric functions arranged in the order before explained; the outside or top line contains the combinations of the same weight arranged as follows, viz. the partitions taken in order from right to left are respectively conjugate to the partitions in the outside column, taken in order from top to bottom; in other words, each square of the sinister diagonal corresponds to two partitions which are conjugate to each other. It is to be noticed that the combinations taken in order, from left to right, are not in the order in which they would be obtained by Arbogast's Method of Derivations from an operand a^n , a being ultimately replaced by unity. The squares above the sinister diagonal are empty (i.e. the coefficients are zero), the greater part of them in virtue of both restrictions, and the remainder in virtue of the second restriction; the empty squares below the sinister diagonal are empty in virtue of the second restriction; but the property was not assumed in the calculation.

¹The notion of Conjugate Partitions is, I believe, due to Professor Sylvester or Mr Ferrers. [It was due to Mr now Dr Ferrers.]

The greater part of the numbers in the Tables (a) were calculated, those of each table from the numbers in the next preceding table by the following method, depending on the derivation of the expression for $b^{p+1}c^q \dots$ from the expression for $b^pc^q \dots$. Suppose, for example, the column cd of Table V (a) is known, and we wish to calculate the column bcd of Table VI (a). The process is as follows:

Given

2^21	21^3	1^5		
1	3	10		
321	2^3	2^21^2	21^4	1^6
1, 3, 2	3, 6, 12		10, 60	
1	3	3, 8, 22	60	

where the numbers in the last line are the numbers in the column bcd of Table VI (a). The partition 2^21 , considered as containing a part zero, gives, when the parts are successively increased by 1, the partitions 321 , 2^3 , 2^21^2 , in which the indices of the increased part (i.e. the original part plus unity) are 1, 3, 2; these numbers are taken as multipliers of the coefficient 1 of the partition 2^21 , and we thus have the new coefficients 1, 3, 2 of the partitions 321 , 2^3 , 2^21^2 . In like manner the coefficient 3 of the partition 21^3 gives the new coefficients 3, 6, 12 of the partitions 31^2 , 2^21^2 , 21^4 , and the coefficient 10 of the partition 1^5 gives the new coefficients 10, 60 of the partitions 21^4 and 1^6 , and finally, the last line is obtained by addition. The process in fact amounts to the multiplication separately of each term of

$$cd = 1(2^21) + 3(21^3) + 10(1^5)$$

by $b = (1)$. It would perhaps have been proper to employ an analogous rule for the calculation of the combinations $cdef \dots$ not containing b , but instead of doing so I availed myself of the existing Tables (b). But the comparison of the last line of each Table (a) (which as corresponding to a combination b^k was always calculated independently of the Tables (b)) with such last line as calculated from the corresponding Table (b), seems to afford a complete verification of both the Tables; and my process has in fact enabled me to detect several numerical errors in the Tables (b), as given in the English translation of the work above referred to. It is not desirable, as regards facility of calculation and independently of the want of verification, to calculate either set of Tables wholly from the other; the rules for the independent calculation of the Tables (b) are fully and clearly explained in the work referred to, and I have nothing to add upon this subject.

The relation of symmetry, alluded to in the introductory paragraph of the present memoir, exists in each Table of either set, and is as follows: viz. the number in the Table corresponding to any two partitions in the outside column and the outside line respectively, is equal to the number corresponding to the same two partitions in the outside line and the outside column respectively. Or, calling the two partitions P, Q , and writing for shortness, combination (P) for the combination represented by the partition P , and for greater clearness, symmetric function (P) (instead of merely (P)) to denote the symmetric function represented by the partition P , we have the following two theorems, viz.

Theorem. The coefficient in combination (P) of symmetric function (Q) is equal to the coefficient in combination (Q) of symmetric function (P) ;

and conversely,

Theorem. The coefficient in symmetric function (P) of combination (Q) is equal to the coefficient in symmetric function (Q) of combination (P) .

M. Borchardt's formula, before referred to, is given in the 'Monatsbericht' of the Berlin Academy (March 5, 1855)², and may be thus stated; viz. considering the case of n roots, write

$$x^n \cdot \frac{1}{k}(1, b, c, \dots | 1, x)^n = (1 - \alpha x)(1 - \beta x) \dots (1 - \kappa x) = fx,$$

then

$$\Sigma \left(\frac{1}{1 - \alpha x} \cdot \frac{1}{1 - \beta y} \cdots \frac{1}{1 - \kappa u} \right) = \frac{1}{k^n} \cdot \frac{fx \, fy \dots fu}{\Pi(x, y, \dots u)} \cdot \frac{d}{dx} \frac{d}{dy} \cdots \frac{d}{du} \frac{\Pi(x, y, \dots u)}{fx \, fy \dots fu},$$

²And in Crelle, t. LIII, p. 195.—Note added 4th Dec. 1857, A. C.

where $\Pi(x, y, \dots u)$ denotes the product of the differences of the quantities $x, y, \dots u$, and on the left-hand side the summation extends to all the different permutations of $\alpha, \beta, \dots \kappa$, or what is the same thing, of $x, y, \dots u$.

Suppose for a moment that there are only two roots, so that

$$(1, b, c \mid 1, x)^2 = (1 - \alpha x)(1 - \beta x),$$

then the left-hand side is

$$\frac{1}{(1 - \alpha x)(1 - \beta y)} + \frac{1}{(1 - \alpha y)(1 - \beta x)},$$

which is equal to

$$2 + (\alpha + \beta)(x + y) + (\alpha^2 + \beta^2)(x^2 + y^2) + 2\alpha\beta xy \\ + (\alpha^3 + \beta^3)(x^3 + y^3) + (\alpha^2\beta + \alpha\beta^2)(x^2y + xy^2) + \&c.,$$

and the right-hand side is

$$\frac{1}{c} \cdot \frac{fx \, fy}{x - y} \cdot \frac{d}{dx} \frac{d}{dy} \frac{x - y}{fx \, fy},$$

which is equal to

$$\frac{1}{c} \cdot \frac{1}{x - y} \left\{ \frac{f'x \, fy - f'y \, fx}{(fx)(fy)} + (x - y) \frac{f'x \, f'y}{fx \, fy} \right\},$$

and therefore to

$$\frac{1}{c} \cdot \frac{1}{fx \, fy} \left\{ \frac{f'x \, fy - f'y \, fx}{x - y} + f'x \, f'y \right\};$$

or substituting for fx, fy their values,

$$\frac{f'x \, fy - f'y \, fx}{x - y}$$

becomes equal to

$$2c + bc(x + y) + 2c^2xy,$$

and $fx \, fy$ is equal to

$$b^2 + 2bc(x + y) + 4c^2xy.$$

The right-hand side is therefore equal to

$$\frac{2 + b(x + y) + 2cxy}{(1 + bx + cx^2)(1 + by + cy^2)},$$

and comparing with the value of the left-hand side, we see that this expression may be considered as the generating function of the symmetric functions of (α, β) , viz. the expression in question is developable in a series of the symmetric functions of (x, y) , the coefficients being of course functions of b and c , and these coefficients are (to given numerical factors *près*) the symmetric functions of the roots (α, β) .

And in general it is easy to see that the left-hand side of M. Borchardt's formula is equal to

$$[0] + 1(1)' + 2(2)' + 1^2(1^2)' + \&c.,$$

where $(1), (2), (1^2), \&c.$ are the symmetric functions of the roots $(\alpha, \beta, \dots \kappa)$, $(1)', (2)', (1^2)', \&c.$ are the corresponding symmetric functions of $(x, y, \dots u)$, and $[0], [1], [2], [1^2], \&c.$ are mere numerical coefficients; viz. $[0]$ is equal to $1 \cdot 2 \cdot 3 \dots n$, and $[1], [2], [1^2], \&c.$ are such that the product of one of these factors into the number of terms in the corresponding symmetric function $(1), (2), (1^2), \&c.$ may be equal to $1 \cdot 2 \cdot 3 \dots n$. The right-hand side of M. Borchardt's formula is therefore, as in the particular case, the generating function of the symmetric functions of the roots $(\alpha, \beta, \dots \kappa)$, and if a convenient expression of such right-hand side could be obtained, we might by means of it express all the symmetric functions of the roots in terms of the coefficients.

The outside line of letters contains the combinations (powers and products) of the coefficients, the coefficients being all with the positive sign, and the coefficient of the highest power being unity; thus in the case of a cubic equation the equation is

$$x^3 + bx^2 + cx + d = (x - \alpha)(x - \beta)(x - \gamma) = 0.$$

The outside line of numbers is obtained from that of letters merely by writing $1, 2, 3 \dots$ for $b, c, d \dots$, and may be considered simply as a different notation for the combinations. The outside column contains the different symmetric functions in the notation above explained, viz. (1) denotes $\Sigma\alpha$, (2) denotes $\Sigma\alpha^2$, (1^2) denotes $\Sigma\alpha\beta$, and so on. The Tables (a) are to be read according to the columns; thus Table II (a) means $b^2 = 1(2) + 2(1^2)$, $c = (1^2)$. The Tables (b) are to be read according to the lines; thus Table II (b) means $(2) = -2c + 1b^2$, $(1^2) = +1c$.

I (a). I (b). (weight 1)

	1 b		1 b
(1)	−1	(1)	−1

II (a). II (b). (weight 2)

	2 c	1 ² b ²		2 c	1 ² b ²
(2)	+1	+2	(2)	−2	+1
(1 ²)		+1	(1 ²)	+1	

III (a). III (b). (weight 3)

	3 d	12 bc	1 ³ b ³		3 d	12 bc	1 ³ b ³
(3)	−1	−3	−1	(3)	−3	+3	−1
(21)		−1	−3	(21)	+3	−1	
(1 ³)			−1	(1 ³)	−1		

IV (a). (weight 4)

	4 e	13 bd	2 ² c ²	1 ² 2 b ² c	1 ⁴ b ⁴
(4)	+1	+4	+2	+6	+24
(31)		+1	+2	+5	+12
(2 ²)			+1	+2	+6
(21 ²)				+1	+4
(1 ⁴)					+1

IV (b). (weight 4)

	4 e	13 bd	2 ² c ²	1 ² 2 b ² c	1 ⁴ b ⁴
(4)	−4	+4	+2	−4	+1
(31)	+4	−1	−2	+1	
(2 ²)	+2		+1		
(21 ²)	−4	+1			
(1 ⁴)	+1				

[The memoir continues with Tables V (a)/(b) (weight 5, columns indexed by the conjugate coefficient-products $f, be, cd, b^2d, bc^2, b^3c, b^5$), VI (a)/(b) (weight 6: $g, bf, ce, b^2e, d^2, bcd, b^3d, c^3, b^2c^2, b^4c, b^6$), VII (a)/(b) (weight 7, occupying journal p. 421), VIII (a)/(b) (weight 8, journal pp. 422–423), IX (a)/(b) (weight 9, journal pp. 424–425, running on through several PDF pages), and X (a)/(b) (weight 10, journal pp. 426–428, the largest tables in the memoir, with column heading running from $k, bi, ch, b^2h, dg, b^3g, bdf, b^2cf, b^4f, be^2, \dots$, through $c^5, b^2c^4, b^4c^3, b^6c^2, b^8c, b^{10}$, and row-labels the partitions of 10 from (10) through (1^{10})).]

The body of each Table (a) is a triangular array of signed integers whose entry in row $(l^\lambda m^\mu \dots)$ and column $b^p c^q \dots$ records the coefficient with which the symmetric function $(l^\lambda m^\mu \dots)$ appears in the expansion of the coefficient-product $b^p c^q \dots$; the Tables (b) invert this relation, giving each symmetric function as a signed integer combination of coefficient-products of equal weight. The squares above the sinister diagonal are empty in virtue of the two restrictions stated above. The symmetry theorem stated by the author asserts that the entry in row P , column Q of any Table N (a) is equal to the entry in row Q , column P of the same table, and likewise for the Tables (b).

The full numerical content of Tables IV (a)/(b) through X (a)/(b) occupies PDF pp. 423–434 of the present volume (= journal pp. 407–418 of the 1857 *Philosophical Transactions*). Tables V through X are wide multi-column grids that resist reflow at 11 pt without sacrificing line-by-line readability; the original journal version (*Phil. Trans.* 1857, vol. CXLVII, pp. 489–499, Tables IV (a)–X (b)) remains the authoritative source for the precise numerical values. A faithful machine-readable transcription of these tables forms a useful subsequent project but is not attempted here for reasons of legibility.